

SOC LAB

NETWORK SECURITY INVESTIGATION

Controlled VirtualBox Security Investigation

Investigation ID	INV-001
Source System	Kali Linux — 192.168.56.101
Target System	Metasploitable — 192.168.56.103
Environment	Isolated VirtualBox Lab
Tools	Nmap • Wireshark • Netcat • FTP • Telnet • NFS
Date	August 2026

1. INTRODUCTION

This project is a controlled SOC lab for practicing basic network security investigation. Kali Linux was used as the investigation system and Metasploitable as the intentionally vulnerable target. The investigation covered reconnaissance, exposed services, network traffic, authentication activity and security misconfigurations.

2. OBJECTIVE

The objective was to identify exposed services, analyze selected network traffic, investigate security findings, and convert the observations into simple SOC detection ideas.

3. LAB ENVIRONMENT

Component	Details
Kali Linux	Investigation machine — 192.168.56.101
Metasploitable	Target machine — 192.168.56.103
VirtualBox	Isolated lab networking
Tools	Nmap, Wireshark, Netcat, FTP, Telnet, NFS

4. NETWORK INFORMATION

Source: 192.168.56.101 | Target: 192.168.56.103 | Network: Isolated VirtualBox lab

5. NETWORK RECONNAISSANCE

Nmap was used to identify open TCP ports and services. The scan confirmed that the target was up and exposed multiple services.

```

[kali@kali]~[~/SOC-LAB-2026]
% nmap 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-13 01:31 -0400
Nmap scan report for 192.168.56.103
Host is up (1.3s latency).
Not shown: 983 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
6667/tcp  open  irc
MAC Address: 08:00:27:3B:C8:7E (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 3.89 seconds

```

6. FTP SERVICE INVESTIGATION

An anonymous FTP login was attempted against TCP port 2121. The server rejected the authentication attempt with a 530 Login incorrect response.

```

[kali@kali]~[~/SOC-LAB-2026]
% ftp 192.168.56.103 2121
Connected to 192.168.56.103.
220 ProFTPD 1.3.1 Server (Debian) [::ffff:192.168.56.103]
Name (192.168.56.103:kali): anonymous
331 Password required for anonymous
Password:
530 Login incorrect.
ftp: Login failed
ftp>

```

7. WIRESHARK: ICMP ANALYSIS

ICMP Echo Request and Echo Reply traffic confirmed communication between Kali and Metasploitable.

The image shows a Wireshark packet capture of ICMP traffic. The packet list on the left shows several ICMP Echo (ping) requests and replies. The selected packet (No. 1) is an ICMP Echo (ping) request from 192.168.56.101 to 192.168.56.103. The packet details pane on the right shows the packet structure: Ethernet II, Internet Protocol Version 4, and ICMP Echo (ping) request. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.56.101	192.168.56.103	ICMP	68	Echo (ping) request id=0x0002, seq=9/2345, ttl=64 (request in 1)
2	0.01597572	192.168.56.103	192.168.56.101	ICMP	68	Echo (ping) reply id=0x0002, seq=9/2345, ttl=64 (reply in 1)
3	0.100483840	192.168.56.101	192.168.56.103	ICMP	68	Echo (ping) request id=0x0002, seq=9/2384, ttl=64 (request in 7)
4	1.000374211	192.168.56.101	192.168.56.103	ICMP	68	Echo (ping) request id=0x0002, seq=9/2384, ttl=64 (request in 0)
5	1.004150654	192.168.56.101	192.168.56.103	ICMP	68	Echo (ping) request id=0x0002, seq=10/2568, ttl=64 (request in 12)
6	2.005952968	192.168.56.103	192.168.56.101	ICMP	68	Echo (ping) reply id=0x0002, seq=10/2568, ttl=64 (reply in 11)

Wireshark - Packet 1 - eth0

- Frame 1: 68 bytes on wire (784 bits), 68 bytes captured (784 bits) on interface eth0, id 0
- Ethernet II, Src: PCSysntec_3b:c8:7e (08:00:27:3b:c8:7e), Dst: PCSysntec_3b:c8:7e (08:00:27:3b:c8:7e)
- Destination: PCSysntec_3b:c8:7e (08:00:27:3b:c8:7e)
- Source: PCSysntec_1e:2d:5d (08:00:27:1e:2d:5d)
- Type: IPv4 (0x0008)
- [Stream index: 0]
- Internet Protocol Version 4, Src: 192.168.56.101, Dst: 192.168.56.103
- 0100 = Version: 4
- 0101 .. = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 84
- Identification: 0x0004 (00004)
- 0101 = Flags: 0x2, Don't fragment
- ..0 0000 0000 0000 = Fragment Offset: 0

0000 00 00 27 3b c8 7e 08 00 27 1e 2d 5d 00 00 45 00 E

0010 00 54 65 64 40 00 40 01 e3 27 c9 a8 38 05 c0 a8 E

0020 38 67 08 0a d1 5d 00 02 00 08 70 5b 7d 6a 00 00 X

0030 00 00 ca ff 0b 00 00 00 00 00 10 11 12 13 14 15 X

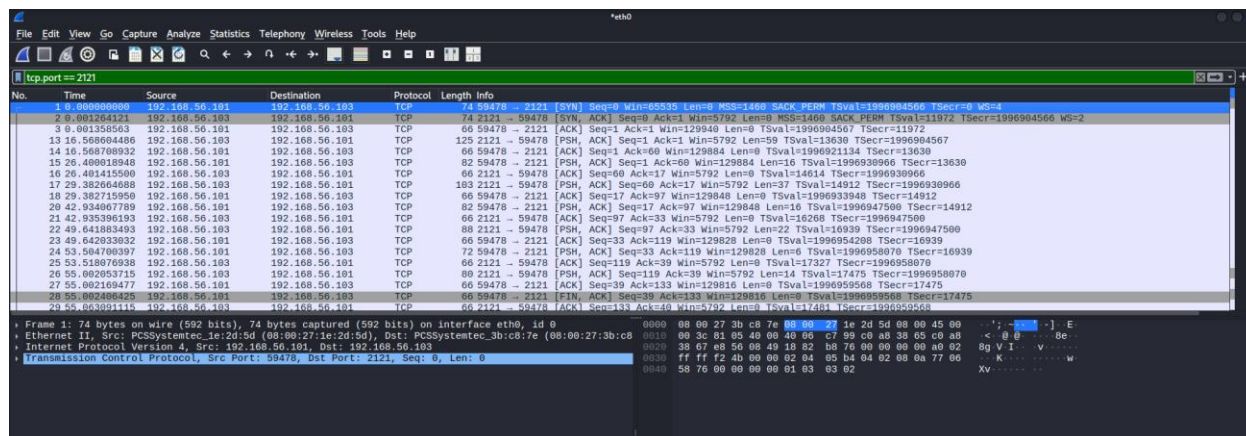
0040 10 17 10 10 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 1

0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 1

0060 36 37 6

8. WIRESHARK: FTP TRAFFIC

FTP-related TCP traffic to port 2121 was captured and reviewed, showing communication between the source and FTP service.



The image shows a Wireshark packet capture of FTP traffic. The filter is set to 'tcp.port==2121'. The packet list shows several TCP packets between 192.168.56.101 and 192.168.56.103. The packet details pane shows the selected packet (No. 74) as a SYN packet from 192.168.56.101 to 192.168.56.103, port 2121. The packet bytes pane shows the raw data of the SYN packet.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.56.101	192.168.56.103	TCP	74	59478 → 2121 [SYN] Seq=0 Win=0 Len=0 MSS=1460 SACK_PERM TSval=1996904566 TSecr=0 WS=2
2	0.001204121	192.168.56.103	192.168.56.101	TCP	74	2121 → 59478 [ACK] Seq=0 Ack=1 Win=0 Len=0 MSS=1460 SACK_PERM TSval=11972 TSecr=1996904566 WS=2
3	0.001358563	192.168.56.101	192.168.56.103	TCP	66	59478 → 2121 [ACK] Seq=1 Ack=1 Win=129948 Len=0 TSval=1996904567 TSecr=11972
13	16.568604486	192.168.56.101	192.168.56.103	TCP	125	2121 → 59478 [PSH, ACK] Seq=1 Ack=1 Win=5792 Len=59 TSval=13630 TSecr=1996904567
14	16.568708932	192.168.56.101	192.168.56.103	TCP	66	59478 → 2121 [ACK] Seq=1 Ack=60 Win=129884 Len=0 TSval=1996921134 TSecr=13630
15	26.409018948	192.168.56.101	192.168.56.103	TCP	82	59478 → 2121 [PSH, ACK] Seq=1 Ack=60 Win=129884 Len=10 TSval=1996930966 TSecr=13630
16	26.401415508	192.168.56.103	192.168.56.101	TCP	66	2121 → 59478 [ACK] Seq=60 Ack=17 Win=5792 Len=0 TSval=14614 TSecr=1996930966
17	29.382604688	192.168.56.103	192.168.56.101	TCP	103	2121 → 59478 [PSH, ACK] Seq=60 Ack=17 Win=5792 Len=37 TSval=14912 TSecr=1996930966
18	29.382715950	192.168.56.101	192.168.56.103	TCP	66	59478 → 2121 [ACK] Seq=17 Ack=97 Win=129848 Len=0 TSval=1996933948 TSecr=14912
20	42.934867789	192.168.56.101	192.168.56.103	TCP	82	59478 → 2121 [PSH, ACK] Seq=17 Ack=97 Win=129848 Len=16 TSval=1996947500 TSecr=14912
21	42.935396193	192.168.56.103	192.168.56.101	TCP	66	2121 → 59478 [ACK] Seq=97 Ack=33 Win=5792 Len=0 TSval=16208 TSecr=1996947500
22	49.641883493	192.168.56.103	192.168.56.101	TCP	88	2121 → 59478 [PSH, ACK] Seq=97 Ack=33 Win=5792 Len=22 TSval=16030 TSecr=1996947500
23	49.642833832	192.168.56.101	192.168.56.103	TCP	66	59478 → 2121 [ACK] Seq=33 Ack=119 Win=129828 Len=0 TSval=1996954208 TSecr=16939
24	53.584709397	192.168.56.103	192.168.56.101	TCP	72	59478 → 2121 [PSH, ACK] Seq=33 Ack=119 Win=129828 Len=6 TSval=1996958076 TSecr=16939
25	53.518076838	192.168.56.103	192.168.56.101	TCP	66	2121 → 59478 [ACK] Seq=119 Ack=39 Win=5792 Len=0 TSval=17327 TSecr=1996958076
26	55.002653715	192.168.56.103	192.168.56.101	TCP	80	2121 → 59478 [PSH, ACK] Seq=119 Ack=39 Win=5792 Len=14 TSval=17475 TSecr=1996958076
27	55.002169477	192.168.56.101	192.168.56.103	TCP	66	59478 → 2121 [ACK] Seq=39 Ack=133 Win=129816 Len=0 TSval=1996959508 TSecr=17475
28	55.002406425	192.168.56.103	192.168.56.101	TCP	66	59478 → 2121 [FIN, ACK] Seq=39 Ack=133 Win=129816 Len=0 TSval=1996959508 TSecr=17475
29	55.003091115	192.168.56.103	192.168.56.101	TCP	66	2121 → 59478 [ACK] Seq=133 Ack=49 Win=5792 Len=0 TSval=17461 TSecr=1996959508

9. TELNET SERVICE

TCP port 23 was reachable and a Telnet connection was established. Telnet should normally be replaced by an encrypted administration protocol such as SSH.

```
[kali@kali]~$ telnet 192.168.56.103 23
Trying 192.168.56.103 ...
Connected to 192.168.56.103.
Escape character is '^]'.
```

10. SMB SERVICE EXPOSURE

TCP ports 139 and 445 were reachable. SMB enumeration later timed out, so this evidence demonstrates service exposure rather than successful enumeration.

```
[kali@kali]~$ nc -vz -w 5 192.168.56.103 445
192.168.56.103: inverse host lookup failed: Unknown host (UNKNOWN) [192.168.56.103] 445 (microsoft-ds) open
[kali@kali]~$ nc -vz -w 5 192.168.56.103 139
192.168.56.103: inverse host lookup failed: Unknown host (UNKNOWN) [192.168.56.103] 139 (netbios-ssn) open
```

11. NFS MISCONFIGURATION

The NFS server exported the root directory to all hosts. The export list displayed '/', representing a serious configuration weakness.

```
[kali@kali]~$ showmount -e 192.168.56.103
Export list for 192.168.56.103:
/ *
```

12. NFS ROOT ACCESS

The export was mounted from Kali and exposed the target filesystem structure, including /etc, /home, /root, /usr and /var.

```
[kali@kali]--[~]
% ls -la ~/nfs-test
total 108
drwxr-xr-x 21 root root 4096 May 20 2012 .
drwx----- 30 kali kali 4096 Aug 14 01:20 ..
drwxr-xr-x 2 root root 4096 May 13 2012 bin
drwxr-xr-x 3 root root 4096 Apr 28 2010 boot
lrwxrwxrwx 1 root root 11 Apr 28 2010 cdrom -> media/cdrom
drwxr-xr-x 2 root root 4096 Apr 28 2010 dev
drwxr-xr-x 94 root root 4096 Aug 13 19:00 etc
drwxr-xr-x 6 root root 4096 Apr 16 2010 home
drwxr-xr-x 2 root root 4096 Mar 16 2010 initrd
lrwxrwxrwx 1 root root 32 Apr 28 2010 initrd.img -> boot/initrd.img-2.6.24-16-server
drwxr-xr-x 13 root root 4096 May 13 2012 lib
drwx----- 2 root root 16384 Mar 16 2010 lost+found
drwxr-xr-x 4 root root 4096 Mar 16 2010 media
drwxr-xr-x 3 root root 4096 Apr 28 2010 mnt
-rw----- 1 root root 10868 Aug 13 19:01 nohup.out
drwxr-xr-x 2 root root 4096 Mar 16 2010 opt
dr-xr-xr-x 2 root root 4096 Apr 28 2010 proc
drwxr-xr-x 13 root root 4096 Aug 13 19:02 root
drwxr-xr-x 2 root root 4096 May 13 2012/sbin
drwxr-xr-x 2 root root 4096 Mar 16 2010/srv
drwxr-xr-x 2 root root 4096 Apr 28 2010/sys
drwxrwxrwt 4 root root 4096 Aug 13 19:19/tmp
drwxr-xr-x 12 root root 4096 Apr 28 2010/usr
drwxr-xr-x 14 root root 4096 Mar 17 2010/var
lrwxrwxrwx 1 root root 29 Apr 28 2010/vmlinuz -> boot/vmlinuz-2.6.24-16-server
```

13. SECURITY FINDINGS

Finding	Risk	Evidence
NFS root filesystem exported to all hosts	High	Figures 7–8
Telnet service exposed	Medium	Figure 5
Multiple network services exposed	Medium	Figure 1
SMB ports 139/445 exposed	Medium	Figure 6
Failed FTP authentication	Info	Figure 2

14. BASIC SOC DETECTIONS

Detection	Detection Logic
Port scanning	One source sends many TCP connection attempts to different ports of one target within a short period.
Failed FTP login	Repeated authentication failures should be correlated with source IP, time and related activity.
Unnecessary service	Unexpected or unrequired services should be identified and reviewed.
NFS misconfiguration	Unrestricted NFS exports, especially sensitive directories such as '/', should generate a high-priority alert.

15. INCIDENT TIMELINE

1. Connectivity confirmed between Kali and Metasploitable. 2. Nmap reconnaissance identified exposed services. 3. FTP authentication was investigated. 4. ICMP and FTP-related traffic were captured in Wireshark. 5. Telnet and SMB exposure were checked. 6. NFS exports were enumerated and the export was mounted. 7. Findings and detection logic were documented.

16. RISK ASSESSMENT

The highest-risk issue was the unrestricted NFS export of the root filesystem because the evidence showed both the configuration and access to the exported filesystem. Telnet and the number of exposed services increase the attack surface and should be reviewed in a real environment.

17. RECOMMENDATIONS

- Restrict NFS exports to trusted hosts only.
- Avoid exporting '/' unless specifically required.
- Disable Telnet and use SSH.
- Disable unnecessary services.
- Restrict SMB access using firewall rules and authentication controls.
- Monitor repeated authentication failures and reconnaissance activity.
- Review service and system logs regularly.

18. CONCLUSION

This investigation demonstrated a basic SOC workflow: establish connectivity, perform reconnaissance, investigate services, analyze traffic, identify findings and develop detection ideas. The strongest finding was the unrestricted NFS root filesystem export. All testing was performed inside the isolated VirtualBox lab.

19. EVIDENCE

Supporting evidence includes Nmap results, FTP authentication output, Wireshark captures, Telnet and SMB connectivity checks, NFS export enumeration, proof of NFS access, PCAP files and investigation notes retained in the project directory.